

Project 4 – Math 3250

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1.A. 0% of people can fit the strap that fits 15 – 21 cm on all 3 of their wrist, biceps, and ankle.

1.B. 94.48% of people can fit the strap that fits 15 – 21 cm on at least 1 of the 3 body parts.

1.C. 11.24% of people can fit the strap that fits 15 – 21 cm on their wrist and either their biceps or ankle.

1.D. The company should market their device (if packaged with the 15 – 21 cm strap) as a device that is primarily meant to be worn as a watch or on the athlete's wrist. Because 94.48% of people can fit that strap on a body part (wrist are smallest) and only 11.24% can fit it on an additional body part.

2.A. 0% of people can fit a strap sized 15 – 19 cm or a strap sized 22 – 32 cm on all 3 body parts (wrists, biceps, and ankle).

2.B. 70.41% of people can fit a strap sized 15 – 19 cm or strap sized 22 - 32 cm on their wrist and at least one other body part (ankle or biceps).

2.C. Everyone ($P = 1$) can fit the strap that is 15 – 19 cm or a strap sized 22 – 32 on some body part (wrist, biceps, or ankles).

2.D. If the company provided 2 straps everyone could fit the strap on at least 1 body part. But only around 70% of people could fit either strap on an additional body part. So, almost 30% of people still would only have a watch or wrist device. It would cost additional money to the company, and it would be difficult to market the device differently sense so many people still cannot use the device in any other configuration.

3.A. No women could wear the 2 included straps of size 14 – 17 cm and 20 – 28 cm on an additional body part to their wrist.

3.B. 1.38% of men could wear the device on their wrist and an additional body part with the included 15 – 19 cm strap and 22 – 36 cm strap included.

3.C. If a strap configuration only works for 95% of people, then it is likely that for every 100 people that buy the product, the configuration will not work for 5 people. For 1000

customers that is 50 people. This continues to scale upward and is problematic to market if an even a small percentage of people are completely unable to use the device in its intended configuration.

3.D. Yes. The shape of our sample should resemble the shape of the data in a larger population. Otherwise, our estimations of percentages are meaningless. The idea of the study is to see what size strap is most likely to fit most people in different configurations. If the data collected in our sample didn't look like the larger population of people, then we would not have correctly estimated what size strap would work best.

3.E. The company should market the device as a wrist or watch device that comes with 2 bands of size 15 – 19 cm and 22 – 32 cm. Some people will be able to use this in other configurations for biceps and ankles, and for those who want those features the company should sell additional band sizes that they can buy. This wouldn't be misleading because it would be marketed primarily as a wrist wearable device.